Database Systems - MCQ Question Bank (Set 6)

50 Multiple Choice Questions | Includes 10 repeated from earlier sets | Answer Key on Last Page

1. Which of the following best defines a Database Management System (DBMS)?

A. A collection of application programs only
B. Software that enables creation, management, and controlled access to a database
C. A type of computer hardware for storing files
D. A programming language used to write operating systems

2. Which key uniquely identifies each record in a table and cannot contain NULL values?

A. Foreign Key
B. Candidate Key
C. Primary Key
D. Alternate Key

3. Which SQL command is used to remove a table's structure along with all its data?

A. DELETE
B. TRUNCATE
C. DROP
D. REMOVE

4. Which of the following is NOT one of the ACID properties of a transaction?

A. Atomicity
B. Consistency
C. Isolation
D. Availability

5. Which indexing structure is most commonly used by databases for efficient range queries?

A. Hash Index
B. B+ Tree Index
C. Bitmap Index
D. Linked List Index

6. A relation is in First Normal Form (1NF) if:

A. It has no transitive dependency
B. All attributes contain only atomic (indivisible) values
C. Every determinant is a candidate key
D. It has no foreign keys

7. Which of the following best describes a deadlock in a DBMS?

A. A query that returns no results
B. Two or more transactions waiting indefinitely for each other to release locks
C. A failure of the database server hardware
D. An error caused by invalid SQL syntax

8. Which SQL clause is used to filter groups after aggregation?

A. WHERE

B. GROUP BY
C. HAVING
D. ORDER BY

9. Which of the following best describes a stored procedure?

A. A precompiled set of SQL statements stored in the database and executed on demand
B. A temporary table used in a query
C. A type of index used for fast lookups
D. A rule enforcing data type constraints

10. Which term describes the property that a completed transaction's changes persist even after a system failure?

A. Atomicity
B. Isolation
C. Durability
D. Consistency

11. Which of the following best describes a superkey?

A. A key that must always be the primary key
B. A set of attributes that uniquely identifies a tuple, possibly with extra attributes
C. A key that never uniquely identifies rows
D. A foreign key referencing multiple tables

12. Which of the following best describes an alternate key?

A. A candidate key not chosen as the primary key
B. A foreign key with no matching primary key
C. A key used only in views
D. A key generated by the operating system

13. Which SQL operator is used to search for a specified pattern in a column?

A. MATCH
B. LIKE
C. FIND
D. SEARCH

14. Which of the following best describes an outer join's purpose?

A. To return only matching rows from both tables
B. To return matching rows plus unmatched rows from one or both tables
C. To remove duplicate rows
D. To sort the result set

15. Which of the following best describes a transitive dependency?

A. A direct dependency of a non-key attribute on the primary key
B. A dependency where a non-key attribute depends on another non-key attribute, which depends on the key
C. A dependency that only exists in denormalized tables
D. A dependency between two primary keys

16. Which of the following best describes the role of a database administrator (DBA)?

A. Writing application-level business logic

B. Managing database security, performance, backup, and recovery
C. Designing the user interface of an application
D. Testing network hardware

17. Which of the following best describes an entity in the ER model?
A. A relationship between two tables
B. A real-world object or concept that can be distinctly identified
C. A constraint on a column
D. A stored procedure

18. Which of the following best describes a derived attribute in an ER diagram?
A. An attribute stored directly in the database
B. An attribute whose value can be calculated from other attributes
C. An attribute that has no data type
D. An attribute used only as a primary key

19. Which of the following best describes a multivalued attribute?
A. An attribute that can hold only a single value
B. An attribute that can hold multiple values for a single entity
C. An attribute that is always a foreign key
D. An attribute that cannot be indexed

20. Which of the following best describes the purpose of the SQL DEFAULT constraint?
A. To specify a value automatically assigned when no value is provided
B. To prevent duplicate values
C. To enforce referential integrity
D. To define the primary key

21. Which of the following best describes the CHECK constraint in SQL?
A. It validates that column values satisfy a specified condition
B. It automatically backs up data
C. It links two tables together
D. It removes duplicate rows

22. Which of the following best describes a recursive relationship in an ER diagram?
A. A relationship between two different entity types
B. A relationship where an entity is related to itself
C. A relationship that cannot have attributes
D. A relationship only used in weak entities

23. Which of the following best describes a data dictionary in a DBMS?
A. A repository of metadata describing the structure of the database
B. A list of all user passwords
C. A backup of all table data
D. A tool for compressing database files

24. Which of the following best describes horizontal partitioning of a table?
A. Splitting columns into separate tables
B. Splitting rows of a table into multiple tables or partitions

C. Merging two unrelated tables
D. Removing all indexes from a table

25. Which SQL function is used to return the current date and time?

A. NOW() or CURRENT_TIMESTAMP
B. DATEVALUE()
C. GETDATA()
D. TIME_NOW()

26. Which of the following best describes a snowflake schema in data warehousing?

A. A schema with a single denormalized fact table only
B. A star schema where dimension tables are further normalized into sub-dimensions
C. A schema used only for transactional systems
D. A schema with no relationships between tables

27. Which of the following best describes the purpose of an ETL process in data warehousing?

A. Encrypt, Transfer, Load data into a single table
B. Extract, Transform, and Load data from source systems into a data warehouse
C. Execute, Test, and Launch database applications
D. Evaluate, Trace, and Log query performance

28. Which of the following best describes conflict serializability?

A. A schedule that can be transformed into a serial schedule by swapping non-conflicting operations
B. A schedule where all transactions run one after another with no overlap
C. A schedule that always causes a deadlock
D. A schedule that ignores read/write conflicts

29. Which of the following best describes timestamp-based concurrency control?

A. Transactions are ordered based on their timestamps to ensure serializability
B. Transactions are executed randomly
C. Only the most recent transaction is allowed to commit
D. Locks are never used

30. Which of the following best describes shadow paging as a recovery technique?

A. Maintaining two page tables, current and shadow, to allow recovery without logging
B. Creating a full backup after every transaction
C. Logging only committed transactions
D. Storing data in encrypted shadow files

31. Which of the following is a common cause of an insertion anomaly?

A. Being unable to add a new record without also supplying data for an unrelated attribute due to poor table design
B. A duplicate primary key value
C. A missing index on a foreign key
D. A corrupted transaction log

32. Which of the following best describes an update anomaly?

A. When updating a value requires changing it in multiple rows due to redundant data, risking inconsistency

- B. When a query fails to execute
- C. When an index becomes corrupted
- D. When a foreign key constraint is violated

33. Which of the following best describes the role of the SQL SELECT INTO statement?

- A. To delete rows from a table
- B. To create a new table and populate it with results from a query
- C. To update existing rows
- D. To grant access privileges

34. Which of the following best describes a database index's main drawback?

- A. It has no drawbacks
- B. It can slow down INSERT, UPDATE, and DELETE operations due to index maintenance
- C. It always increases query execution time
- D. It removes the need for primary keys

35. Which of the following best describes the term 'normal form' in database design?

- A. A set of guidelines used to reduce redundancy and improve data integrity in relational databases
- B. A backup format for exporting data
- C. A programming language for writing queries
- D. A type of database index

36. Which of the following best describes a domain constraint in a relational database?

- A. A rule restricting the values an attribute can take based on its data type and range
- B. A rule linking two tables together
- C. A rule that defines table names
- D. A rule that manages user sessions

37. Which of the following best describes the purpose of the SQL EXISTS operator?

- A. To check whether a subquery returns any rows
- B. To count the number of rows in a table
- C. To create a new index
- D. To sort query results

38. Which of the following best describes a correlated subquery?

- A. A subquery that references a column from the outer query and is re-evaluated for each outer row
- B. A subquery that runs completely independently of the outer query
- C. A subquery that cannot use WHERE clauses
- D. A subquery used only in DELETE statements

39. Which of the following best describes database concurrency?

- A. Preventing any two transactions from ever running at the same time
- B. Allowing multiple transactions to execute simultaneously while maintaining data consistency
- C. Running all queries sequentially with no parallelism
- D. Encrypting data during transmission

40. Which of the following best describes the purpose of database backup and recovery mechanisms?

- A. To improve query performance

B. To protect against data loss and restore the database after a failure
C. To enforce user access control
D. To normalize database tables

41. Which of the following best describes a full backup versus an incremental backup?

A. A full backup copies all data; an incremental backup copies only data changed since the last backup
B. They are identical in function
C. An incremental backup always takes longer than a full backup
D. A full backup only copies the transaction log

42. Which of the following best describes a database view's benefit for security?

A. It can restrict users to seeing only specific columns or rows of underlying tables
B. It encrypts the entire database
C. It prevents all users from querying data
D. It automatically deletes sensitive data

43. Which of the following best describes the purpose of the SQL BETWEEN operator?

A. To select values within a specified inclusive range
B. To join two tables
C. To remove duplicate rows
D. To grant user permissions

44. Which of the following best describes referential actions like ON DELETE CASCADE?

A. Automatically deleting or updating related rows in child tables when a parent row is deleted or updated
B. Preventing any deletion in the database
C. Encrypting deleted rows
D. Logging all SELECT queries

45. Which of the following best describes a database's physical schema?

A. The way data is logically organized for users
B. The actual storage details, such as file organization and indexing on disk
C. The set of user permissions
D. The list of SQL queries used by an application

46. Which of the following best describes the three-schema architecture's conceptual level?

A. Describes what data is stored and the relationships among it, independent of physical storage
B. Describes how individual users view the data
C. Describes the physical storage structures only
D. Describes the network protocol used to access the database

47. Which of the following best describes the external level in the three-schema architecture?

A. Describes physical storage details
B. Describes how different user groups view a subset of the database
C. Describes the entire database schema for all users identically
D. Describes the backup schedule

48. Which of the following best describes a foreign key that allows NULL values?

A. The relationship it represents is optional for that row
B. It always causes a constraint violation
C. It cannot be used in relational databases
D. It automatically becomes a primary key

49. Which of the following best describes the term 'query throughput' in database performance?

A. The number of queries a system can process in a given time period
B. The size of the database file
C. The number of tables in a schema
D. The number of users registered in the system

50. Which of the following best describes column-oriented (columnar) storage in databases?

A. Data is stored row by row for transactional workloads
B. Data is stored column by column, which benefits analytical queries scanning few columns
C. Data is stored only in memory, never on disk
D. Data is stored as unstructured text files

Answer Key

1. B 2. C 3. C 4. D
5. B 6. B 7. B 8. C
9. A 10. C 11. B 12. A
13. B 14. B 15. B 16. B
17. B 18. B 19. B 20. A
21. A 22. B 23. A 24. B
25. A 26. B 27. B 28. A
29. A 30. A 31. A 32. A
33. B 34. B 35. A 36. A
37. A 38. A 39. B 40. B
41. A 42. A 43. A 44. A
45. B 46. A 47. B 48. A
49. A 50. B